Sahil **Kureel**

About me

Atal Bihari Vajpayee - Indian Institute of Information Technology & Management, Gwalior - (B.TECH + M.TECH) In Information Technology

Software Engineer offering 5+ years of experience building scalable distributed platforms and cloud-native architectures. Specialized in Python, NodeJs, Java, and microservices design, with deep domain expertise in payment systems, KYC process and event-driven infrastructure. Proven track record in designing scalable architecture, optimizing cloud-native applications, and implementing secure, resilient systems. Passionate about solving complex architectural challenges and delivering robust, production-grade software.

Contact Info

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Programming Languages

- Python 3+ • Java 17+
- JavaScript (ES6+)
- SQL • Bash

Frameworks Libraries

Frameworks:
- FastAPI | Flask | Django (Python)
- NodeJs | ReactJs | ExpressJs | Tailwind (Javascript)
- spring-boot | Maven (Java)

Databases and Storages: PostgreSQL / AuroraDB (RDS) | DynamoDB | S3 / GCS / Firehose | MongoDB | Redis / Memcached
Cloud Tech: AWS | GCP | LocalStack
Messaging Systems: RabbitMQ | Kafka | SQS
Tools: Docker | Jenkins-pipline | SonarQube | Git | Terraform
Concepts:
Linux Systems | Distributed Systems | System Designs at High Level as well as Low Level | Performance Optimization & Tuning for scale | Data Structures & Algorithms | REST API Designs Principle (SOLID) | Concurrency Control using Mutex / Semaphores for Multithreading and Asynchronous Programming | Encryption at REST using AES-CBC / AES-GCM

AI Data Tools and Tech

- **Ollama / GitHub Copilot / GPT-4 / Claude Haiku 4.5**
- **DynamoDB-Streams / PySpark / Apache-Airflow**
- **NumPy / Pandas / SciPy**
- **LLMs Integration with Ollama & FastAPI Interface**

Experiences

Aug'21 – Sept'25 | **HEAD DIGITAL WORKS PVT. LTD.**
SDE - III (Platform Team) · Bengaluru
- Designed and developed scalable microservices using Spring-Boot, FastAPI, ExpressJs and Django / django-rest-framework for Payment Processing, KYC Verification, Encryption at REST and Bank Account Validation.
- Built a Unified Payment System integrated Juspay and Razorpay payment-Aggregator capable of handling high-volume traffic for financial transactions (100M/day & 1M TPM).
- Architected event-driven systems using AWS ECS / Lambda, API Gateway, DynamoDB Streams, S3 Streams and RabbitMQ.
- Owned production systems and resolved critical reliability and performance issues in distributed environments.
- Implemented secure and scalable cloud-native infrastructure using AWS WAF, Lambda / ECS, and Serverless Framework.
- Led code reviews, architectural discussions and CI/CD implementation using Jenkins with SonarQube like Quality Gates.
- Collaborated with Client Application teams in integrating backend APIs, along with ReactJS-based dashboards designs and internal tracking tools.

Tech: Python(FastAPI / Django), NodeJS(ExpressJs), Java(Spring-Boot), AWS, RabbitMQ & Kafka, DynamoDB, RDS, Docker, ReactJs, Tailwind-CSS

Aug'20–July'21 | **GARG CONSULTANCY CO.**
Software Engineer - Backend · Gurugram
- Led backend development for financial and digital platforms using Python and NodeJs.
- Designed and implemented RESTful APIs, CMS systems and backend services with scalability and security best practices.
- Collaborated with product and clients to deliver production-ready solutions.
- Mentored junior developers and improved engineering workflows.
- Managed cloud deployments and backend infrastructure on AWS.
- Supported frontend integrations with ReactJS & NextJs applications.

Tech: Python(Flask/Django), NodeJs(ExpressJs), AWS, PostgreSQL, RabbitMQ, Docker

May'19–Mar'20 | **LOCALWELL**
Software Engineer - Backend · Ahmedabad
- Developed backend services for a SaaS-based pharma supply chain platform.
- Built APIs and services, designed database schemas, and optimized performance.
- Delivered MVP backend architecture from scratch.

Tech: NodeJs - TypeScipt (ECMAScript) / Strapi CMS, Flask / Django, PostgreSQL, MongoDB, Docker

Projects

July 2023 **Reconciliation System**
Distributed Financial Reconciliation Engine ·
• Designed and implemented a distributed reconciliation engine using DynamoDB Streams, processing 500,000+ daily transactions to resolve discrepancies across multiple ledger systems.
• Built high-performance interface using lambda for reconciliation triggers, audit reporting, and transaction status tracking.
• Implemented asynchronous processing and multithreading for handling large-scale transaction workloads efficiently includes concurrency control.
• Developed fault-tolerant workflows with retries, logging, scheduled jobs, and automated reporting.
• Deployed services on AWS lambda / ECS with auto-scaling support for variable workloads.
• Discrepancy Resolution flagging 99.9% of financial leaks automatically.
• Manual Effort saving 15+ hours of weekly manual operations.
• Concurrency handling 2,000+ concurrent requests per second.
• **Tech Stack:** Python3, NodeJs, Java21, RabbitMQ, PostgreSQL, AWS Lambda / ECS, Docker.

May 2022 **Bank Account Management System**
Banking Backend Platform ·
• Designed and implemented a scalable banking backend using Flask with middleware-based authentication, auditing, and request validation.
• Built transactional ledger systems using AWS RDS with ACID-compliant transaction management.
• Used MongoDB for operational data management and RabbitMQ for asynchronous synchronization workflows.
• Implemented scalable APIs and backend services for account management and financial operations.
• **Tech Stack:** Python, Flask, MongoDB, AWS RDS, RabbitMQ, Docker

May 2020-2024 **AI-Based Tourism Recommendation System**
Based on Indian Scenario ·
• Architected an intelligent tourism platform using Django and MongoDB, integrating TensorFlow and Scikit-learn to deliver personalized, multi-destination travel itineraries.
• Engineered collaborative filtering and content-based recommendation models that improved user engagement and itinerary acceptance rates.
• Built scalable pipelines using Apache Spark and AWS Redshift to aggregate and process GiBs of real-time travel history and destination metrics.
• Deployed the containerized application on AWS Elastic Beanstalk, configuring automated scaling to maintain sub-second latency for concurrent active users.
• High Performance: 92% recommendation accuracy score / 25% boost in itinerary acceptance.
• **Tech Stack:** Python3, FastAPI, AWS Aurora (PostgreSQL), MongoDB, RabbitMQ, Apache Kafka, Redis, Snowflake, Apache Airflow, Docker

Certificates & Grants

2016 SnakDown Qualifier Round-3, CodeChef Coding Competition.
2017 Management Team (Technical Assistant) at ACM - ICPC, Gwalior region.
2019 AWS Certified Solution Architect Associate.